Math 63: Real Analysis

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Problem 1. Show that a complete subspace of a metric space is a closed subset.

Solution.

Let S be a complete subspace of a metric space (E, d). By the proposition on page 51, we know that every convergent sequence of points in a metric space is a Cauchy sequence, and moreover by defition of a complete space we know that every cauchy sequence of points $p \in S$ converges to a point of S. From the claim in Problem (4.2) we know that if every convergenet sequence of points in S converges to a point in S, S is a closed set. Hence, a complete subspace of a metric space is a closed subset.

Problem 2. Let $a, b \in \mathbb{R}, a < b$. The following outlines a proof that $[a, b]$ is compact. Rewrite this proof, filling i n all details: Let $\{U_i\}_{i \in I}$ be a collection of open subsets of $\mathbb{R}$ whose union contains $[a, b]$. Let $S = \{x \in][a, b] : x > a$ and $[a, x)$ is contained in the union of a finite number of the sets $\{U_i\}_{i \in I}\}$. Then $l.u.b.S \in U_i$ for some $i \in I$. Since U_i is open we must have $l.u.b.S = b \in S$.

Problem 3. Call a metric space sequentially compact if every sequence has a convergent subsequence. Prove that a metric space is sequentially compact if and only if every infinite subset has a cluster point.

Problem 4. Call a metric space totally bounded if, for every $\epsilon > 0$, the metric space is the union of a finite number of closed balls of radius ϵ. Prove that a metric space is totally bounded if and only if every sequence has a Cauchy subsequence.

Problem 5. Prove that the following three conditions on a metric space E (cf. Probs. 35, 36) are equivalent:

1. E is compact.
2. E is sequentially convergenet.
3. E is totally bounded and complete.

(Hint: That (1) implies (2) occurs in the text. That (2) implies (3) is easy. That (3) implies (1) follows from the argument of the last proof of § 5.)